

Tugas Mobile Programming Pertemuan 3

Nama : Lucas Gabriel Mahatan

NIM : 20240801012

Github : https://github.com/LucasssssPakeC/Lucas-Gabriel-Mahatan_Kelompok-4_kh002.git

BAGIAN 1

1. Analisis Kategori Widget

Klasifikasi Widget:

- Standalone Widget: Text, FlutterLogo (*Widget Tampilan Dasar yang tidak menerima properti anak child maupun children*).
- Single-Child Widget: Container, Center (*Widget pembungkus/tata letak yang hanya menampung satu widget anak melalui properti child*).
- Multiple-Children Widget: Column, Row (*Widget struktur tata letak yang menampung daftar/banyak widget anak melalui properti children*).

Perbedaan Properti child vs children:

- child: Menerima satu objek Widget tunggal. Digunakan pada *Single-Child Widget* untuk memodifikasi, memberi batasan, atau membungkus satu elemen.
- children: Menerima daftar widget (`List<Widget>`). Digunakan pada *Multiple-Children Widget* untuk menyusun beberapa widget sekaligus secara sejajar (vertikal atau horizontal).

2. Analisis Skenario Error Layout (Unbounded Constraints)

- Masalah Visual (Error UI Banner):

Layar emulator akan menampilkan error "A RenderFlex overflowed by [X] pixels" yang ditandai secara visual dengan munculnya banner/pita garis-garis kuning dan hitam (*yellow and black striped banner*) di bagian tepi kanan layar tempat teks terpotong.

- Widget Solusi (Single-Child) & Korelasi Modul:

Widget yang digunakan adalah Expanded (*kategori Single-Child Widget*). Membungkus widget Text dengan Expanded akan memaksa teks tersebut mengambil sisa ruang horizontal yang tersedia pada Row secara fleksibel. Konsep ini sepadan dengan atribut `android:layout_weight="1"` pada LinearLayout Android Native (sebagaimana dijelaskan pada modul halaman 7–8), di mana elemen diberikan bobot (*weight*) untuk mengisi seluruh sisa ruang kosong wadahnya.

3. Perilaku Sumbu Utama (MainAxisAlignment)

Pada Column dengan 3 widget anak (di mana sumbu utamanya adalah vertikal):

- MainAxisAlignment.center:

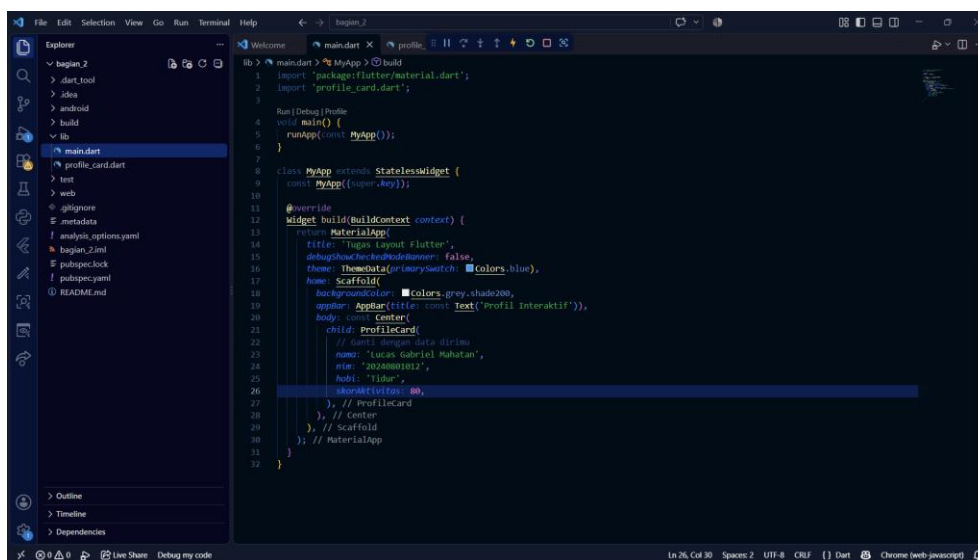
Ketiga widget anak akan diposisikan berkumpul rapat di tengah-tengah sumbu vertikal Column. Seluruh sisa ruang kosong dialokasikan di bagian atas widget pertama dan di bawah widget ketiga, tanpa ada jarak tambahan di antara masing-masing widget.

- `MainAxisAlignment.spaceEvenly`:

Seluruh sisa ruang kosong dibagikan secara sama rata di sepanjang sumbu vertikal. Akibatnya, jarak antara batas atas ke widget pertama, jarak antar-widget (1 ke 2, dan 2 ke 3), serta jarak dari widget ketiga ke batas bawah memiliki ukuran piksel yang persis sama.

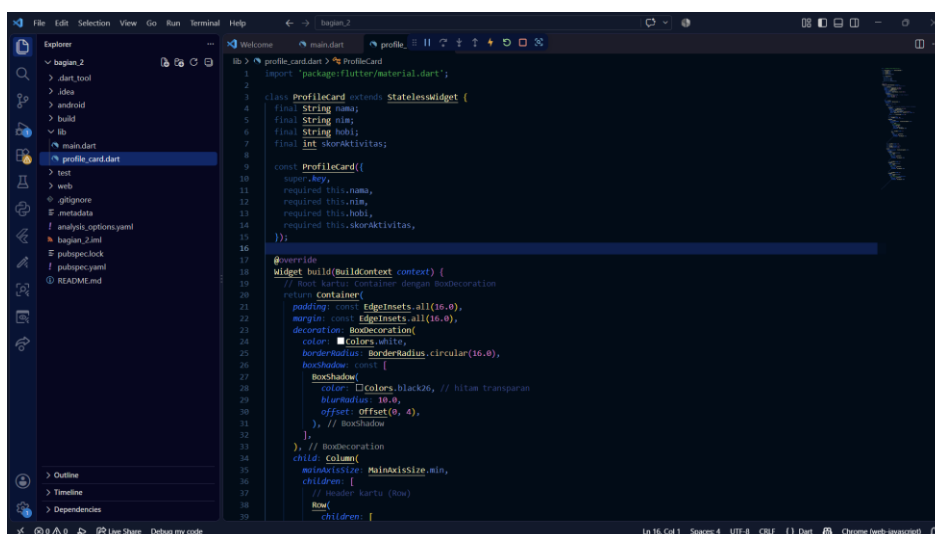
BAGIAN 2

Main.dart

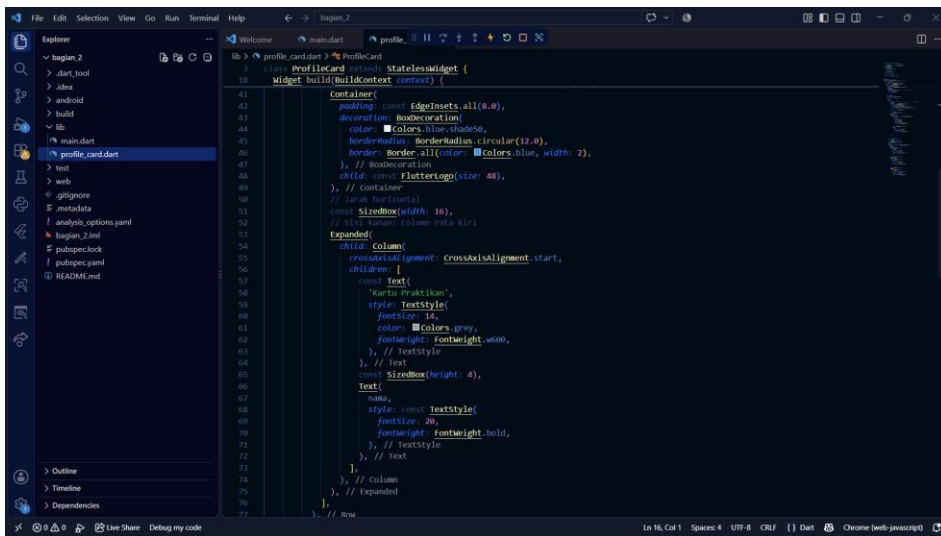


```
1 import 'package:flutter/material.dart';
2 import 'profile_card.dart';
3
4 void main() {
5   runApp(MyApp());
6 }
7
8 class MyApp extends StatelessWidget {
9   const MyApp({super.key});
10
11   @override
12   Widget build(BuildContext context) {
13     return MaterialApp(
14       title: 'Tugas Layout Flutter',
15       debugShowCheckedStackBanner: false,
16       theme: ThemeData(primarySwatch: Colors.blue),
17       home: Scaffold(
18         backgroundColor: Colors.grey.shade200,
19         appBar: AppBar(title: const Text('Profil Interaktif')),
20         body: const Center(
21           child: ProfileCard(
22             // data: data dari
23             nama: 'Lucas Gabriel Mahatan',
24             nmr: '2024001012',
25             hobi: 'Futsal',
26             skorAktivitas: 80,
27           ), // ProfileCard
28         ), // Center
29       ), // Scaffold
30     ); // MaterialApp
31   }
32 }
```

Profile_card.dart

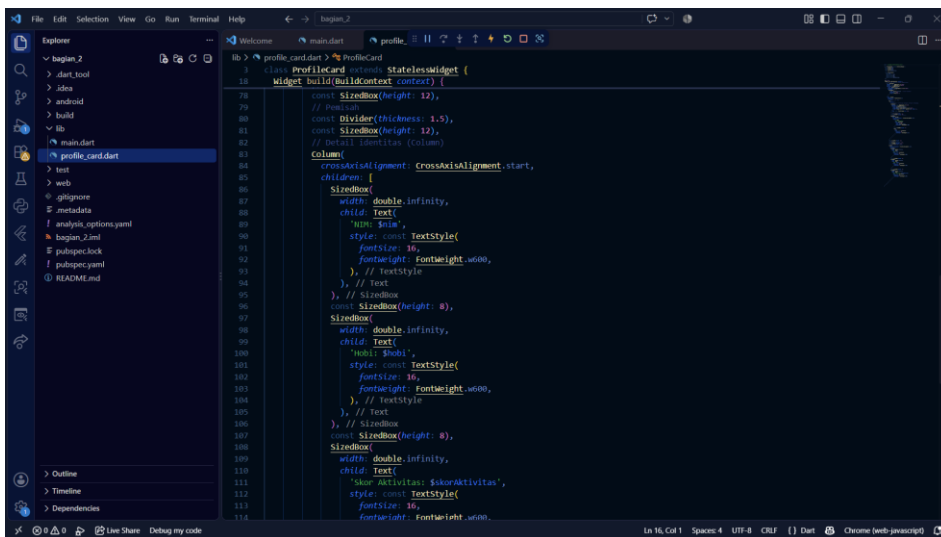


```
1 import 'package:flutter/material.dart';
2
3 class ProfileCard extends StatelessWidget {
4   final String nama;
5   final String nmr;
6   final String hobi;
7   final int skorAktivitas;
8
9   const ProfileCard({
10     super.key,
11     required this.nama,
12     required this.nmr,
13     required this.hobi,
14     required this.skorAktivitas,
15   });
16
17   @override
18   Widget build(BuildContext context) {
19     // Card akan Container dengan BoxDecoration
20     return Container(
21       padding: const EdgeInsets.all(16.0),
22       margin: const EdgeInsets.all(16.0),
23       decoration: BoxDecoration(
24         color: Colors.white,
25         borderRadius: BorderRadius.circular(16.0),
26         boxShadow: [
27           BoxShadow(
28             color: Colors.black26, // hitam transparan
29             blurRadius: 16.0,
30             offset: Offset(0, 4),
31           ), // BoxShadow
32         ],
33       ), // BoxDecoration
34       child: Column(
35         mainAxisAlignment: MainAxisAlignment.min,
36         children: [
37           // header kartu (Row)
38           Row(
39             children: [
40               // ...
41             ],
42           ),
43         ],
44       ),
45     );
46 }
```



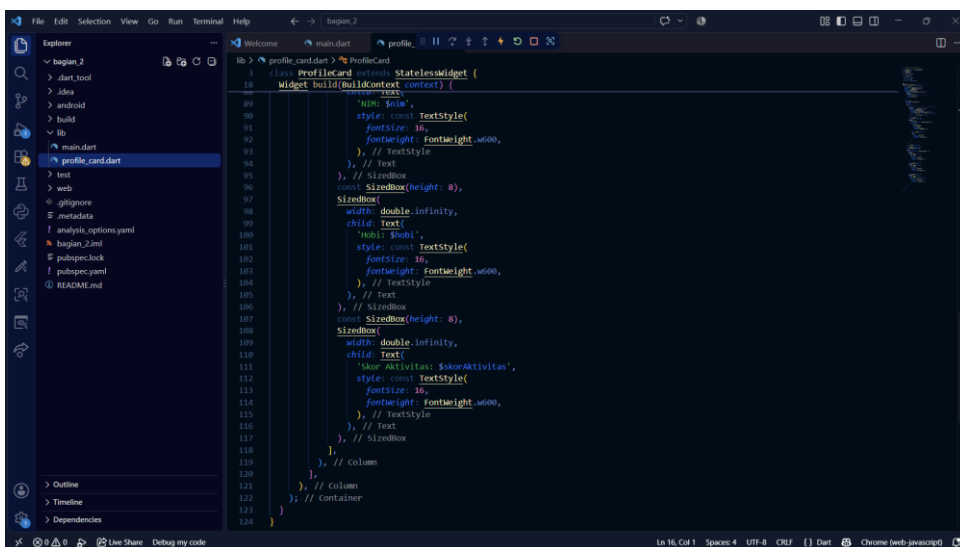
The screenshot shows an IDE with the Dart file `profile_card.dart` open. The code defines a `ProfileCard` widget that extends `StatelessWidget`. It includes a `build` method that returns a `Widget` tree. The widget has a blue background with a white border and a circular logo. The text "Kartu Praktikan" is displayed in a bold, black font. The code is as follows:

```
1 class ProfileCard extends StatelessWidget {
2   Widget build(BuildContext context) {
3     Container(
4       padding: const EdgeInsets.all(8.0),
5       decoration: BoxDecoration(
6         color: Colors.blue.shade50,
7         borderRadius: BorderRadius.circular(12.0),
8         border: Border.all(color: Colors.blue, width: 2),
9       ), // BoxDecoration
10      child: const FlutterLogo(size: 48),
11    ), // Container
12    // Container (optional)
13    const SizedBox(width: 16),
14    // Sisi Kanan: kolom rata kiri
15  ), // Expanded
16  child: Column(
17    crossAxisAlignment: CrossAxisAlignment.start,
18    children: [
19      const Text(
20        'Kartu Praktikan',
21        style: TextStyle(
22          fontSize: 16,
23          color: Colors.grey,
24          fontWeight: FontWeight.w600,
25        ), // TextStyle
26      ), // Text
27      const Text(
28        'Nama: $nama',
29        style: TextStyle(
30          fontSize: 20,
31          fontWeight: FontWeight.bold,
32        ), // TextStyle
33      ), // Text
34    ], // Column
35  ), // Expanded
36  ), // Row
37 }
```



The screenshot shows the same IDE with the `profile_card.dart` file. The code has been updated to include a `Divider` widget and a `Column` widget. The widget now has a white background with a blue border and a blue logo. The text "Kartu Praktikan" is displayed in a bold, black font. The code is as follows:

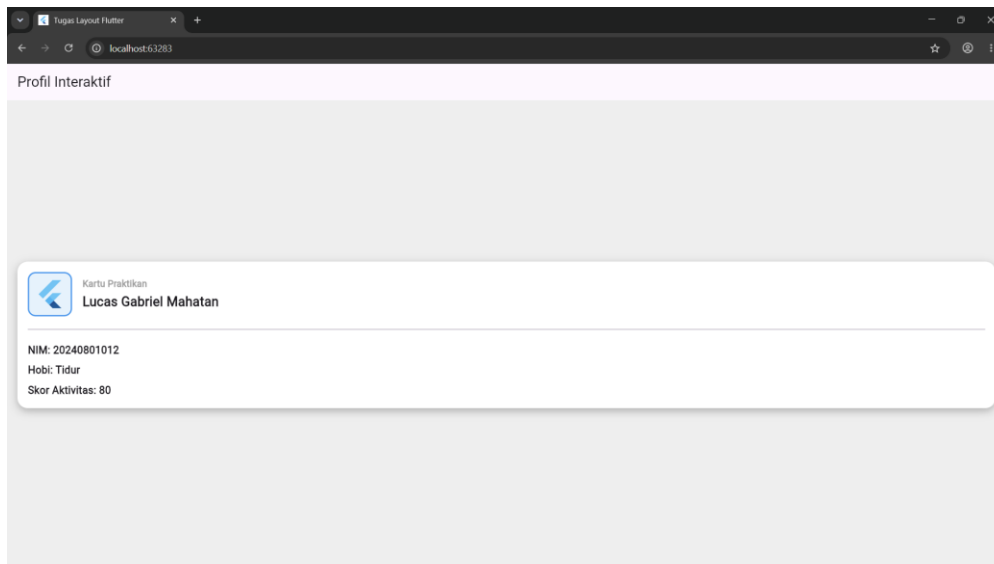
```
76 class ProfileCard extends StatelessWidget {
77   Widget build(BuildContext context) {
78     // Profile Card
79     // BoxDecoration
80     const Divider(thickness: 1.5),
81     const SizedBox(height: 12),
82     // Divider (optional)
83     Column(
84       crossAxisAlignment: CrossAxisAlignment.start,
85       children: [
86         const Text(
87           'Kartu Praktikan',
88           style: TextStyle(
89             fontSize: 16,
90             fontWeight: FontWeight.w600,
91           ), // TextStyle
92         ), // Text
93         const Text(
94           'Nama: $nama',
95           style: TextStyle(
96             fontSize: 20,
97             fontWeight: FontWeight.bold,
98           ), // TextStyle
99         ), // Text
100        const Text(
101          'Show Aktivitas: $showAktivitas',
102          style: TextStyle(
103            fontSize: 16,
104            fontWeight: FontWeight.w600,
105          ), // TextStyle
106        ), // Text
107      ], // Column
108    ), // Expanded
109  ), // Row
110 }
```



The screenshot shows the same IDE with the `profile_card.dart` file. The code has been updated to include a `Column` widget and a `Text` widget. The widget now has a white background with a blue border and a blue logo. The text "Kartu Praktikan" is displayed in a bold, black font. The code is as follows:

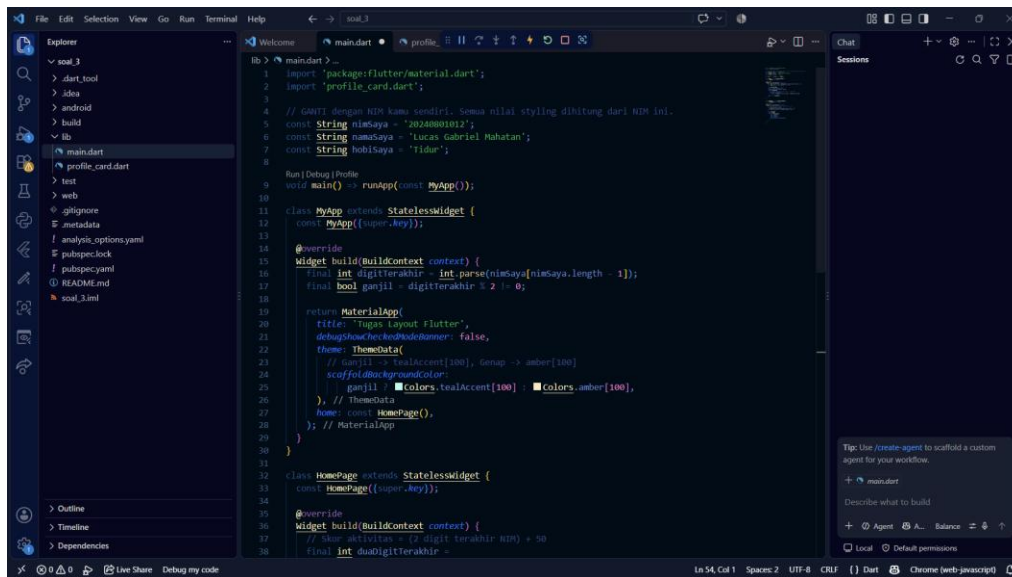
```
89 class ProfileCard extends StatelessWidget {
90   Widget build(BuildContext context) {
91     // Profile Card
92     // BoxDecoration
93     const Text(
94       'Kartu Praktikan',
95       style: TextStyle(
96         fontSize: 16,
97         fontWeight: FontWeight.w600,
98       ), // TextStyle
99     ), // Text
100    const Text(
101      'Nama: $nama',
102      style: TextStyle(
103        fontSize: 20,
104        fontWeight: FontWeight.bold,
105      ), // TextStyle
106    ), // Text
107    const Text(
108      'Show Aktivitas: $showAktivitas',
109      style: TextStyle(
110        fontSize: 16,
111        fontWeight: FontWeight.w600,
112      ), // TextStyle
113    ), // Text
114  ], // Column
115 ), // Expanded
116 ), // Row
117 }
```

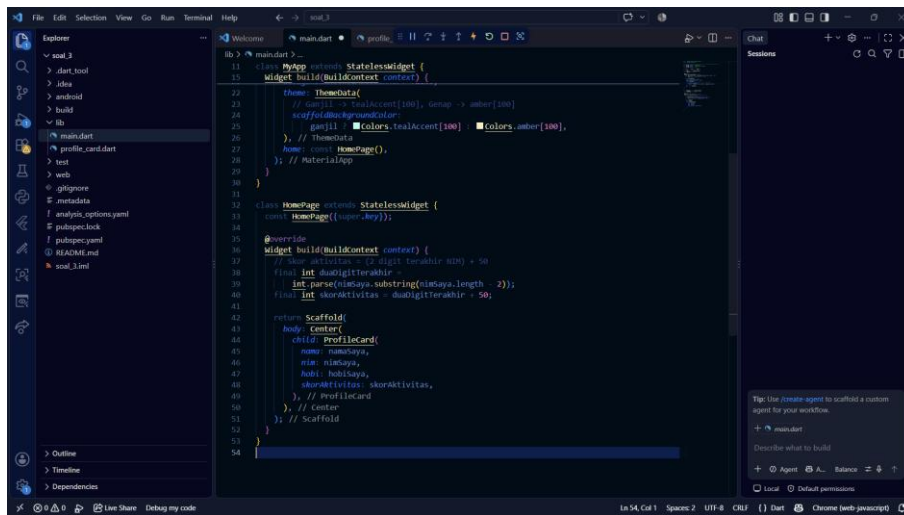
Tampilan Chrome Web



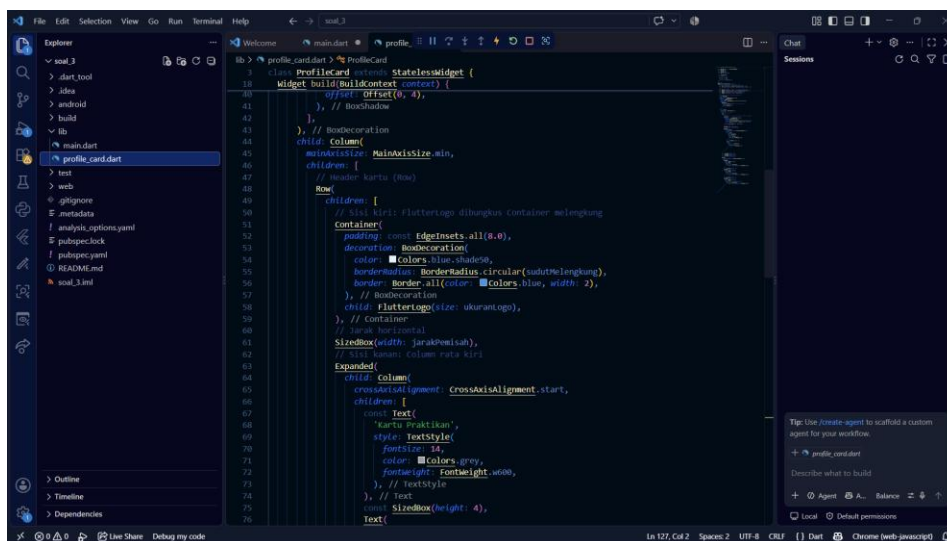
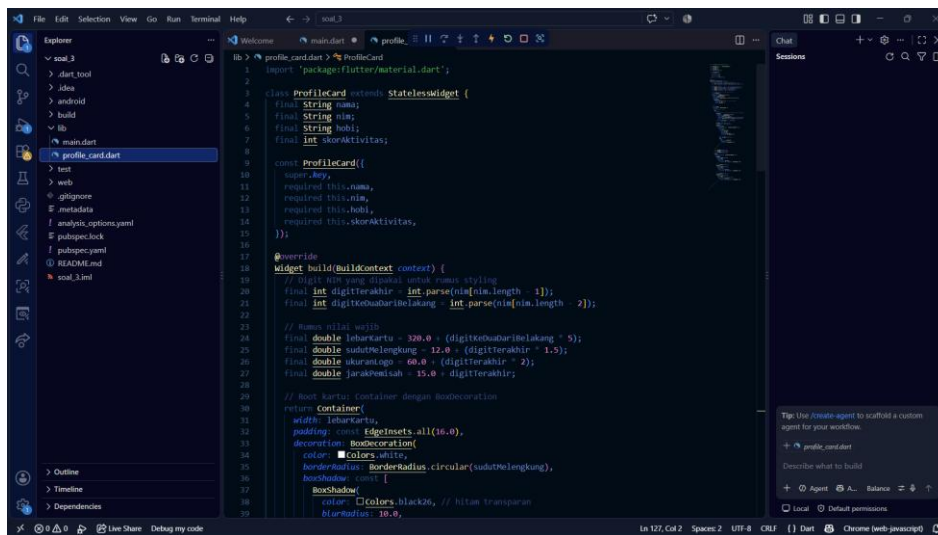
BAGIAN 3

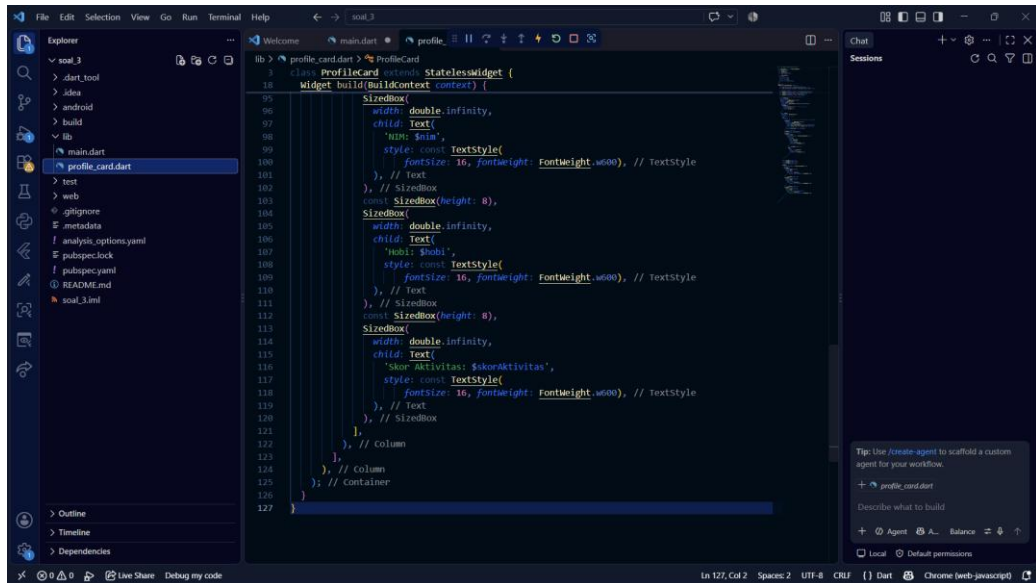
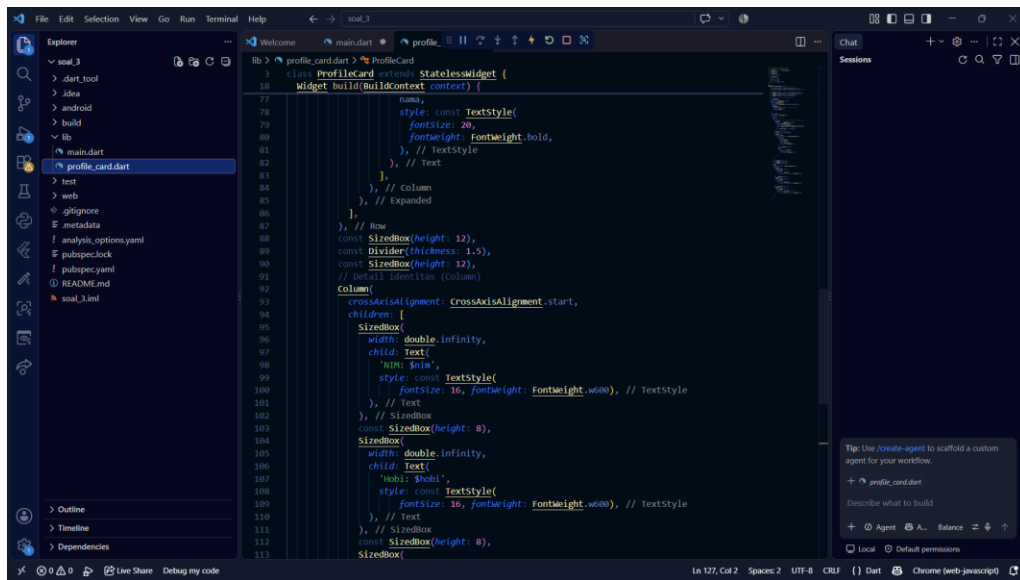
Main.dart





Profile_card.dart





Tampilan Chrome web

